# Practice Worksheet 1: Computation

CORE 103 Computational Thinking

Fall 2026

## Instructions

- This PW is UNGRADED. However, it is highly encouraged that you attempt and submit it to get feedback from your section tutor.
- This PW must be submitted online via CANVAS. You are required to submit a zip file that contains all the *.py* files.
- The zip file should be named as *PW1_aa123456.zip* where *aa123456* will be replaced with your student id.

## Exercises

### 1. Campus Booth Shift

*This question should be implemented in the file* ***q1.py****.*

During a campus event, volunteers sign up for booth duty using same-day p.m. times from 1 p.m. to 11 p.m. One student starts helping at $S$ p.m. and leaves when the shift ends at $E$ p.m.

The event team wants to record the length of each volunteer's shift for its attendance sheet.

Given $S$ and $E$, determine the number of hours in the volunteer's shift.

**Constraints**

- $S, E \in \mathbb{N}$
- $1 \leq S < E \leq 11$

**Interaction**

The input comprises a single line containing 2 space-separated integers denoting the values of $S$ and $E$ respectively.

The output must contain a single number denoting the volunteer's shift duration in hours.

**Sample**

| Input | Output |
|---|---|
| 4 6 | 2 |
| 1 11 | 10 |

In the first case, $(S, E) = (4, 6)$. The campus event booth is active from 4 p.m. to 6 p.m. Therefore, the student volunteers for 2 hours.

In the second case, $(S, E) = (1, 11)$. The campus event booth is active from 1 p.m. to 11 p.m. Therefore, the student volunteers for 10 hours.

**Exercise**

Indicate the outputs for the given inputs in **q1.py**.

| Input | Output |
|---|---|
| 1 3 | |
| 3 7 | |
| 9 11 | |

**Pseudocode**

Implement the pseudocode in **q1.py**.

## 2. Biryani Workshop Budget

*This question should be implemented in the file **q2.py**.*

A university cooking society is planning a short biryani workshop for its members. The culinary instructor charges Rs. $F$ for each class, and the society has already decided how many classes it will attend each week.

The workshop will run for $W$ weeks, with $C$ classes scheduled every week. The society treasurer needs the complete amount before collecting money from members.

Given $F$, $C$, and $W$, determine the total amount of money the society has to pay.

**Constraints**

- $F, C, W \in \mathbb{Z}$
- $1 \leq F \leq 5000$
- $1 \leq W \leq 52$
- $1 \leq C \leq 7$

**Interaction**

The input comprises a single line containing 3 space-separated integers denoting the values of $F$, $C$, and $W$ respectively.

The output must contain a single number denoting the total amount of money the society has to pay.

**Sample**

| Input | Output |
|---|---|
| 500 3 3 | 4500 |
| 1000 7 1 | 7000 |

In the first case, $(F, C, W) = (500, 3, 3)$. The instructor charges 500 rupees per class, and the society attends a total of 9 classes (3 classes per week for 3 weeks). The society has to pay $500 \cdot 9 = 4500$ rupees.

In the second case, $(F, C, W) = (1000, 7, 1)$. The instructor charges 1000 rupees per class, and the society attends a total of 7 classes (7 classes per week for 1 week). The society has to pay $1000 \cdot 7 = 7000$ rupees.

**Exercise**

Indicate the outputs for the given inputs in **q2.py**.

| Input | Output |
|---|---|
| 1350 5 4 | |
| 500 7 52 | |
| 1337 3 10 | |

**Pseudocode**
Implement the pseudocode in **q2.py**.

## 3. Spreadsheet Expansion

*This question should be implemented in the file **q3.py**.*

A student is maintaining a class society registration spreadsheet after orientation week. The sheet initially contains $R$ rows and $C$ columns.

Later, $E$ more students sign up, so the student adds $E$ extra rows while keeping the number of columns the same. The student now wants to know how many cells the updated sheet contains.

Given $R$, $C$, and $E$, calculate the new total number of cells.

**Constraints**

- $R, C, E \in \mathbb{N}$
- $1 \leq R \leq 100$
- $1 \leq C \leq 100$
- $1 \leq E \leq 100$

**Interaction**

The input comprises a single line containing 3 space-separated integers denoting the values of $R$, $C$ and $E$ respectively.

The output must contain a single number denoting the total number of cells in the sheet.
**Sample**

| Input | Output |
|---|---|
| 5 2 1 | 12 |
| 6 10 3 | 90 |

In the first case, $(R, C, E) = (5, 2, 1)$. Initially there were 5 rows and 2 columns, so the total number of cells was $5 \times 2 = 10$. Adding one more row makes 6 rows; therefore, the new total is $6 \times 2 = 12$.

In the second case, $(R, C, E) = (6, 10, 3)$. Initially there were 6 rows and 10 columns, so the total number of cells was $6 \times 10 = 60$. Adding three more rows makes 9 rows; therefore, the new total is $9 \times 10 = 90$.

**Exercise**

Indicate the outputs for the given inputs in **q3.py**.

| Input | Output |
|---|---|
| 7 3 2 | |
| 3 19 8 | |
| 17 5 6 | |

**Pseudocode**
Implement the pseudocode in **q3.py**.

## 4. Campus Store Discount

*This question should be implemented in the file **q4.py**.*

During a campus sale, the Habib University store applies a percentage discount to one item at the billing counter. The original price of the item is stored as `cp` rupees, and the discount rate is stored as `dr` percent.

The cashier's system must show the final amount to be paid after applying the discount.

Given `cp` and `dr`, determine the sale price of the item after the discount is applied.

**Constraints**

- `cp` and `dr` are integers.
- $0 \leq$ `cp` $\leq 100000$
- $0 \leq$ `dr` $\leq 100$

**Interaction**

The input comprises a single line containing 2 space-separated integers denoting the values of `cp` and `dr`, respectively.

The output must contain a floating-point value denoting the sale price of the item in rupees.

**Sample**

| Input | Output |
|---|---|
| 200 10 | 180.0 |
| 100 20 | 80.0 |

In the first case, the original price is 200 rupees and the discount rate is 10 percent. The discount is 20 rupees, so the sale price is 180.0 rupees.

In the second case, the original price is 100 rupees and the discount rate is 20 percent. The discount is 20 rupees, so the sale price is 80.0 rupees.

**Exercise**

Indicate the outputs for the given inputs in **q4.py**.

| Input | Output |
|---|---|
| 500 0 | |
| 85 10 | |
| 50 100 | |

**Pseudocode**

Implement the pseudocode in **q4.py**.

## 5. University Trip Capacity

*This question should be implemented in the file **q5.py**.*

A university society is planning a one-day trip and has arranged two types of vans. Some vans have 5 seats, while the larger vans have 7 seats.

The organizers have $F$ vans of the first type and $S$ vans of the second type. Before opening registration, they need to know the maximum number of people who can travel together.

Given $F$ and $S$, determine the maximum number of people who can travel together.

**Constraints**

- $F, S \in \mathbb{Z}$
- $0 \leq F, S \leq 100$

**Interaction**

The input comprises a single line containing 2 space-separated integers denoting the values of $F$ and $S$ - the number of 5-seat vans and 7-seat vans, respectively.

The output must contain a single number denoting the maximum number of people that can travel together in these vans.

**Sample**

| Input | Output |
|---|---|
| 4 8 | 76 |
| 2 13 | 101 |

In the first case, $(F, S) = (4, 8)$. There are 4 vans that seat 5 each and 8 vans that seat 7 each. So, $(4 \times 5) + (8 \times 7) = 76$ people can travel together.

In the second case, $(F, S) = (2, 13)$. There are 2 vans that seat 5 each and 13 vans that seat 7 each. So, $(2 \times 5) + (13 \times 7) = 101$ people can travel together.

**Exercise**

Indicate the outputs for the given inputs in **q5.py**.

| Input | Output |
|---|---|
| 50 2 | |
| 8 68 | |
| 9 23 | |

**Pseudocode**

Implement the pseudocode in **q5.py**.

## 6. Library Late Fine Check

*This question should be implemented in the file **q6.py**.*

A university library is checking overdue books for a small student group. The library charges Rs. $F$ for each overdue book.

There are $T$ students in the group. Each student had at most one book to return, and $R$ of the students returned their books on time. The librarian wants to calculate the total fine collected from the remaining students.

Given $F$, $T$, and $R$, calculate the total fine collected.

**Constraints**

- $F, T, R \in \mathbb{Z}$
- $1 \leq F \leq 10$
- $0 \leq R \leq T \leq 10$

**Interaction**

The input comprises a single line containing 3 space-separated integers denoting the values of $F$, $T$ and $R$ respectively.

The output must contain a single number denoting the total fine collected by the library.

**Sample**

| Input | Output |
|---|---|
| 4 1 1 | 0 |
| 2 10 7 | 6 |

In the first case, $(F, T, R) = (4, 1, 1)$. The total number of people with an overdue book is $1 - 1 = 0$. The total fine collected is $4 \times 0 = 0$.

In the second case, $(F, T, R) = (2, 10, 7)$. The total number of students with an overdue book is $10 - 7 = 3$. The total fine collected is $3 \times 2 = 6$.

**Exercise**

Indicate the outputs for the given inputs in **q6.py**.

| Input | Output |
|---|---|
| 8 5 4 | |
| 9 7 0 | |
| 4 5 2 | |

**Pseudocode**

Implement the pseudocode in **q6.py**.

## 7. Gaming Energy

*This question should be implemented in the file **q7.py**.*

In a mobile game, a player's character has $E$ energy points before starting a level. A special power move can be used multiple times, but each use consumes $C$ energy points.

The game only allows a power move if the character has enough energy for the complete move. The player wants to know how many complete power moves are possible.

Given $C$ and $E$, determine the maximum number of complete power moves the character can perform.

**Constraints**

- $C, E \in \mathbb{N}$
- $1 \leq C \leq 100$
- $1 \leq E \leq 1000$

**Interaction**

The input comprises a single line containing 2 space-separated integers denoting the values of $C$ and $E$ respectively.

The output must contain a single number denoting the maximum number of power moves that the character can perform.

**Sample**

| Input | Output |
|---|---|
| 10 30 | 3 |
| 6 41 | 6 |

In the first case, $(C, E) = (10, 30)$. The character can perform a maximum of 3 power moves, which use 30 energy points.

In the second case, $(C, E) = (6, 41)$. The character can perform a maximum of 6 power moves, which use 36 energy points, with 5 energy points to spare.

**Exercise**

Indicate the outputs for the given inputs in **q7.py**.

| Input | Output |
|---|---|
| 50 2 | |
| 8 68 | |
| 9 23 | |

**Pseudocode**

Implement the pseudocode in **q7.py**.

## 8. Hostel Photo Wall Clips

*This question should be implemented in the file **q8.py**.*

A university student is decorating their hostel room with a photo string containing pictures of friends, family, and campus memories. The string is $L$ centimeters long.

Photo clips are attached every $D$ centimeters, starting exactly at position 0 and ending exactly at position $L$. The length of the string is chosen so that the clips fit evenly, with no position skipped.

Given $L$ and $D$, determine the total number of photo clips required.

**Constraints**

- $L, D \in \mathbb{Z}$
- $1 \leq D \leq L \leq 1000$
- $L$ is divisible by $D$

**Interaction**

The input comprises a single line containing 2 space-separated integers denoting the values of $L$ and $D$ respectively.

The output must contain a single number denoting the total number of photo clips required.

**Sample**

| Input | Output |
|---|---|
| 24 6 | 5 |
| 35 7 | 6 |

In the first case, $(L, D) = (24, 6)$. The clips are placed at positions 0, 6, 12, 18, and 24. Therefore, a total of 5 photo clips are required.

In the second case, $(L, D) = (35, 7)$. The clips are placed at positions 0, 7, 14, 21, 28, and 35. Therefore, a total of 6 photo clips are required.

**Exercise**

Indicate the outputs for the given inputs in **q8.py**.

| Input | Output |
|---|---|
| 18 3 | |
| 48 8 | |
| 63 9 | |

**Pseudocode**
Implement the pseudocode in **q8.py**.

## 9. Paired Programming Partner

*This question should be implemented in the file **q9.py**.*

A programming class is arranging students for a paired activity. There are $P$ pairs in total, so the roll numbers go from 1 to $2P$.

The instructor pairs students from opposite ends of the roll-number list: roll number 1 with roll number $2P$, roll number 2 with roll number $2P - 1$, and so on.

Given your roll number $R$, determine your partner's roll number.

**Constraints**

- $P, R \in \mathbb{Z}$
- $1 \leq P \leq 300$
- $1 \leq R \leq 2P$

**Interaction**

The input comprises a single line containing 2 space-separated integers denoting the values of $P$ and $R$, respectively.

The output must contain a single number denoting the roll number of your partner.

**Sample**

| Input | Output |
|---|---|
| 2 2 | 3 |
| 3 1 | 6 |

In the first case, $(P, R) = (2, 2)$. There are $2 \cdot 2 = 4$ students in the class and your roll number is 2. The pairs will be as follows: $\{(1, 4), (2, 3)\}$. You are partnered with roll number 3.

In the second case, $(P, R) = (3, 1)$. There are $2 \cdot 3 = 6$ students in the class and your roll number is 1. The pairs will be as follows: $\{(1, 6), (2, 5), (3, 4)\}$. You are partnered with roll number 6.

**Exercise**

Indicate the outputs for the given inputs in **q9.py**.

| Input | Output |
|---|---|
| 15 5 | |
| 35 23 | |
| 123 189 | |

**Pseudocode**
Implement the pseudocode in **q9.py**.